Project Workflow

The system consists of four primary roles:

- Admin
- Customer
- Agent
- Claims Officer

The application implements Role-Based Access Control (RBAC) using JWT authentication.

SYSTEM ENTRY FLOW (All Roles)

User clicks "Login"

→ System validates credentials (JWT generated)
→ Role identified
→ Redirect based on role:

- Admin → Admin Dashboard
- Customer → Customer Dashboard
- Agent → Agent Dashboard
- Claims Officer → Claims Dashboard

ADMIN WORKFLOW (Click-Based Flow)

User Management

Admin clicks "Manage Users"
→ System shows list of users
→ Admin clicks "Create Agent"
→ Fill form → Submit
→ Agent record saved → Success message

Admin clicks "Create Claims Officer"
→ Fill form → Submit
→ Saved → Success

Admin clicks "Edit User"
→ Modify details → Save

Admin clicks "Delete Customer"
→ System checks:

- If no policies → Delete

- If policies exist → Show error

Assignment Management

Admin clicks “Assign Agent”
→ Select Customer
→ Select Agent
→ Click Assign
→ PolicyAssignment.AgentId updated

Admin clicks “Assign Claims Officer”
→ Select Claim
→ Select Officer
→ Click Assign

Monitoring

Admin clicks “View Policies”
→ List displayed
→ Filter by Status

Admin clicks “View Claims”
→ List displayed
→ Filter by Status

CUSTOMER WORKFLOW (Step-by-Step Flow)

Registration Flow

Click “Register”
→ Fill details (Name, DOB, Email, etc.)
→ Submit
→ Account created
→ Redirect to Plans Page

Policy Purchase Flow

Click “View Plans”
→ Plans displayed

Click “View Details”
→ Plan details page opens

Click “Buy Policy”
→ System asks:

- Add Policy Members
- Add Nominee(s)

Click “Calculate Premium”
→ System calculates premium
→ Premium displayed

Click “Submit Request”
→ PolicyAssignment created
→ Status = Pending
→ Agent notified

Agent Review Flow

Agent clicks “View Assigned Requests”
→ Policy details shown

Agent clicks “Approve”
→ Policy status = Active
→ Customer notified

OR

Agent clicks “Reject”
→ Policy status = Rejected
→ Customer notified

Premium Payment Flow

Customer clicks “Pay Premium”

→ Premium details displayed

→ Select payment method

→ Click “Proceed”

→ Loading buffer

→ Payment success page

→ Payment saved in DB

→ Invoice generated

→ Download PDF option

Premium Reminder Flow (Background Process)

Daily scheduler runs:
→ Check policies where NextDueDate approaching
→ Send email reminder

If payment late:
→ Add 2% interest
→ Show updated payable amount

Claim Filing Flow

Customer clicks “My Policies”
→ Select Policy

Click “File Claim”
→ Select Policy Member
→ Enter claim details
→ Upload documents
→ Submit

→ Claim created
→ Status = Pending
→ Claims Officer notified in order to change status

Claim Processing Flow

Claims Officer clicks “View Assigned Claims”
→ Review details

Click “Approve”
→ Status = Approved

Click “Settle”
→ Status = Settled
→ Policy status updated
→ Nominees notified

OR

Click “Reject”
→ Status = Rejected
→ Remarks added

AGENT WORKFLOW

Login
→ Dashboard

Click “Assigned Policies”
→ View requests and click “Approve / Reject”

Click “Support Tickets”
→ Reply to customer

Dashboard shows:

- Policies sold
- Commission earned
- Performance stats

CLAIMS OFFICER WORKFLOW

Login
→ Claims Dashboard

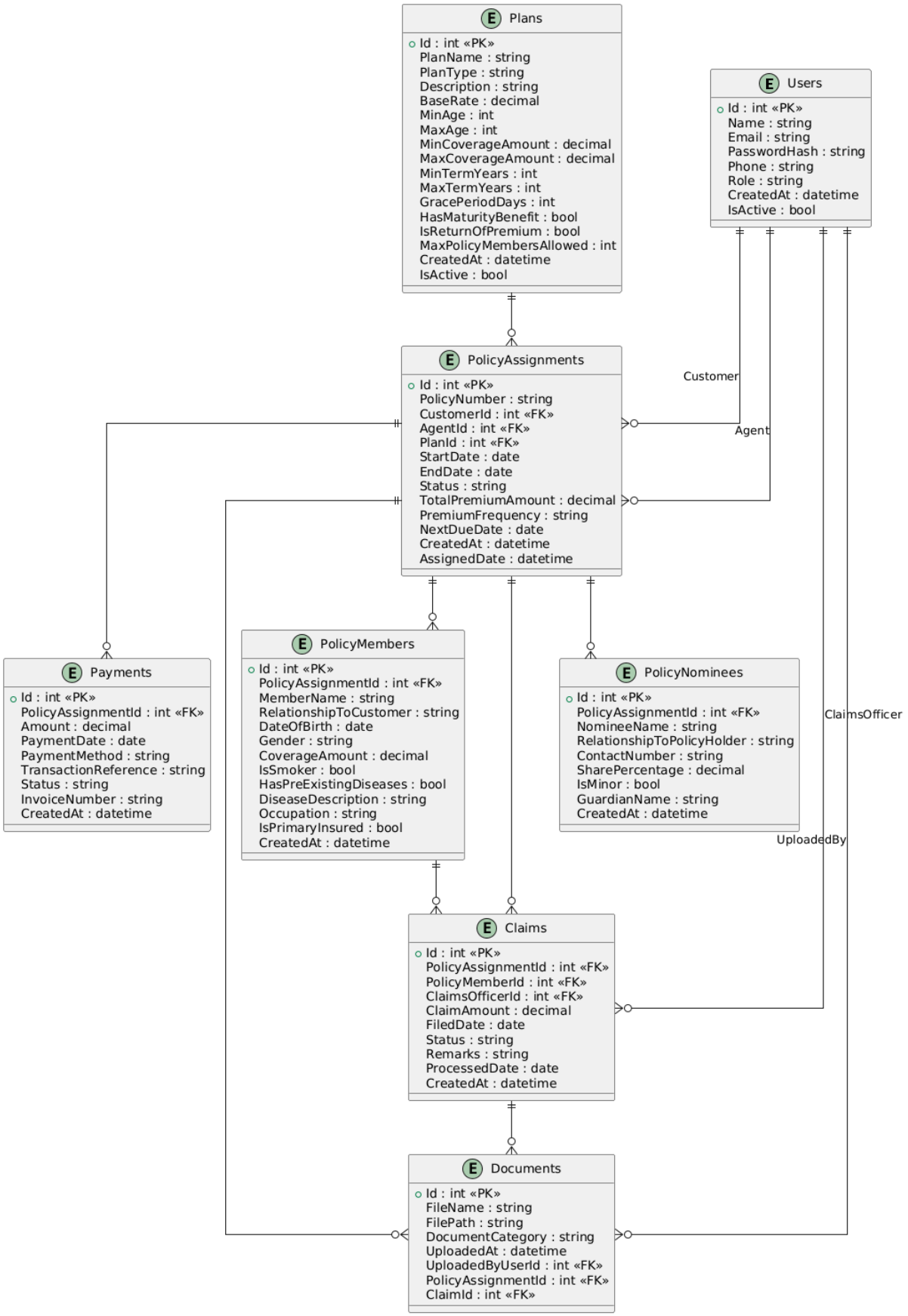
Click “Assigned Claims”
→ Review documents

Click “Approve / Reject / Settle” and update status then the Customer will be notified

Tables

- Users – Id, Name, Email, PasswordHash, Phone, Role, CreatedAt, IsActive
- Plans – Id, PlanName, PlanType, Description, BaseRate, MinAge, MaxAge, MinCoverageAmount, MaxCoverageAmount, MinTermYears, MaxTermYears, GracePeriodDays, HasMaturityBenefit, IsReturnOfPremium, MaxPolicyMembersAllowed, CreatedAt, IsActive
- PolicyAssignments - Id, PolicyNumber, CustomerId, AgentId, PlanId, StartDate, EndDate, Status, TotalPremiumAmount, PremiumFrequency, NextDueDate, CreatedAt, AssignedDate
- PolicyMembers – Id, PolicyAssignmentId, MemberName, RelationshipToCustomer, DateOfBirth, Gender, CoverageAmount, IsSmoker, HasPreExistingDiseases, DiseaseDescription, Occupation, IsPrimaryInsured, CreatedAt
- PolicyNominees – Id, PolicyAssignmentId, NomineeName, RelationshipToPolicyHolder, ContactNumber, SharePercentage, IsMinor, GuardianName, CreatedAt
- Claims – Id, PolicyAssignmentId, PolicyMemberId, ClaimsOfficerId, ClaimAmount, FiledDate, Status, Remarks, ProcessedDate, CreatedAt
- Payments – Id, PolicyAssignmentId, Amount, PaymentDate, PaymentMethod, TransactionReference, Status, InvoiceNumber, CreatedAt
- Documents – Id, FileName, FilePath, DocumentCategory, UploadedAt, UploadedByUserId, PolicyAssignmentId, ClaimId

ER Diagram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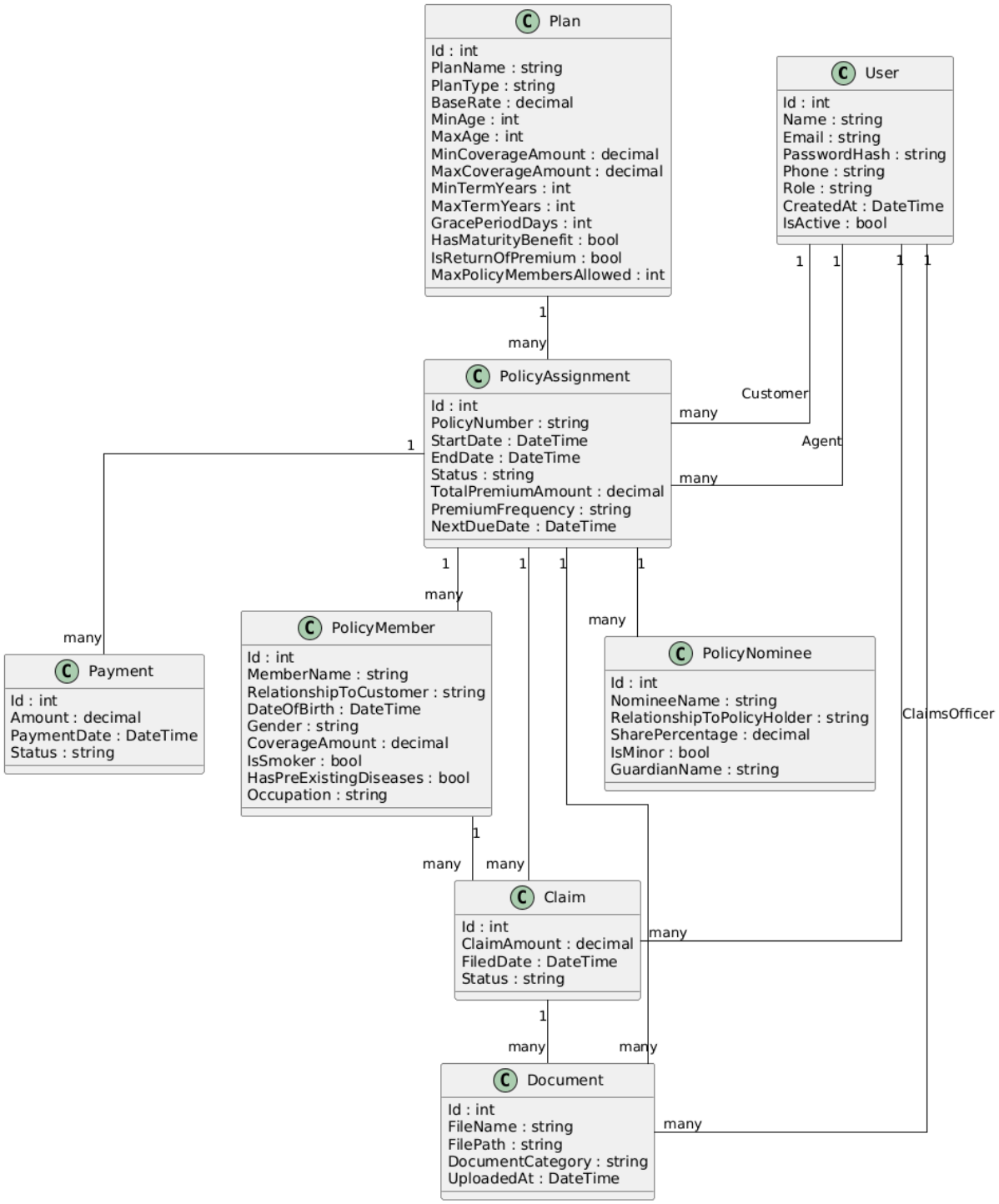
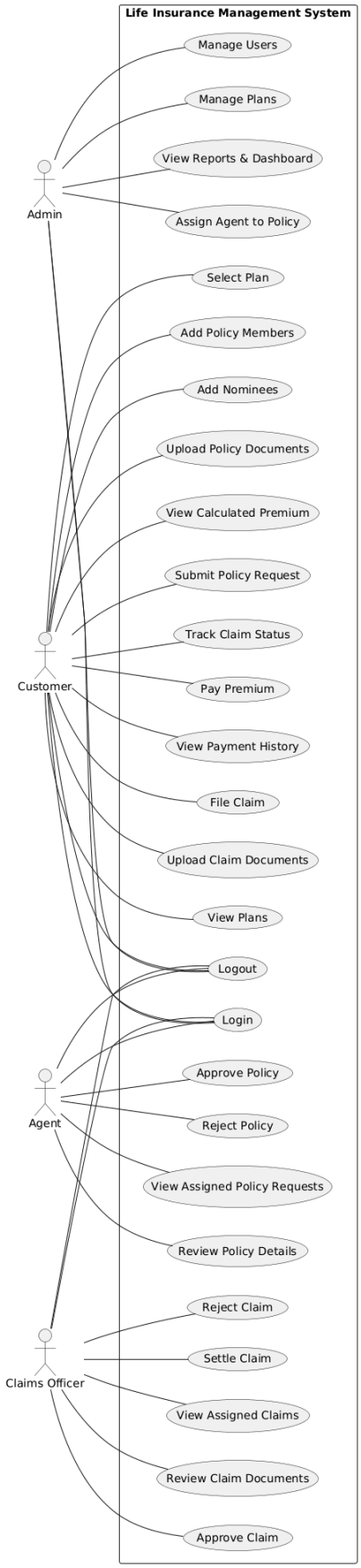
PlanName : string
PlanType : string
Description : string
BaseRate : decimal
MinAge : int
MaxAge : int
MinCoverageAmount : decimal
MaxCoverageAmount : decimal
MinTermYears : int
MaxTermYears : int
GracePeriodDays : int
HasMaturityBenefit : bool
IsReturnOfPremium : bool
MaxPolicyMembersAllowed : int
CreatedAt : datetime
IsActive : bool
Users
Id : int «PK»
Name : string
Email : string
PasswordHash : string
Phone : string
Role : string
CreatedAt : datetime
IsActive : bool
PolicyAssignments
Id : int «PK»
PolicyNumber : string
CustomerId : int «FK»
AgentId : int «FK»
PlanId : int «FK»
StartDate : date
EndDate : date
Status : string
TotalPremiumAmount : decimal
PremiumFrequency : string
NextDueDate : date
CreatedAt : datetime
AssignedDate : datetime
Customer
Agent
ClaimsOfficer
UploadedBy
Payments
Id : int «PK»
PolicyAssignmentId : int «FK»
Amount : decimal
PaymentDate : date
PaymentMethod : string
TransactionReference : string
Status : string
InvoiceNumber : string
CreatedAt : datetime
PolicyMembers
Id : int «PK»
PolicyAssignmentId : int «FK»
MemberName : string
RelationshipToCustomer : string
DateOfBirth : date
Gender : string
CoverageAmount : decimal
IsSmoker : bool
HasPreExistingDiseases : bool
DiseaseDescription : string
Occupation : string
IsPrimaryInsured : bool
CreatedAt : datetime
PolicyNominees
Id : int «PK»
PolicyAssignmentId : int «FK»
NomineeName : string
RelationshipToPolicyHolder : string
ContactNumber : string
SharePercentage : decimal
IsMinor : bool
GuardianName : string
CreatedAt : datetime
Claims
Id : int «PK»
PolicyAssignmentId : int «FK»
PolicyMemberId : int «FK»
ClaimsOfficerId : int «FK»
ClaimAmount : decimal
FiledDate : date
Status : string
Remarks : string
ProcessedDate : date
CreatedAt : datetime
Documents
Id : int «PK»
FileName : string
FilePath : string
DocumentCategory : string
UploadedAt : datetime
UploadedByUserId : int «FK»
PolicyAssignmentId : int «FK»
ClaimId : int «FK»

Class Diagram



Use Case Diagram